Personalized Temporal Modeling of Social Media Addiction Risk

Using Behavioral Change Patterns and Explainable Machine Learning

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1. Introduction and Problem Statement

Social media usage among students has risen significantly over the past few years, and with it, the concern around compulsive usage patterns and digital addiction. Most existing studies rely on one-time surveys or static usage snapshots — for example, a student reports an average of 6 hours of daily screen time — but these single data points miss the bigger picture of how a student's behavior is actually changing over time.

This project takes a different approach. Instead of asking whether a student uses their phone for a lot of hours, we ask: *is this student's screen time increasing relative to their own past behavior?* Are they checking their phone more often than they used to? Is their night-time usage creeping up week over week? By tracking these behavioral shifts across a 4-week observation window, we can catch early signs of escalation before they snowball into a full-blown problem.

The core idea is personalized temporal modeling. Each student serves as their own control. We compute Week 1 usage as their personal baseline and then measure how much their behavior deviates from that baseline over the following weeks. These deviations — what we call delta features — become the primary inputs to an XGBoost machine learning model that predicts addiction risk. We then use SHAP explainability to show exactly which behavioral changes are driving a student's risk score, and we wrap the whole thing in an interactive Dash web dashboard where counselors or researchers can explore the data visually.

2. Objectives

- Simulate realistic longitudinal social media usage data for 800 students across 4 weeks, covering metrics like screen time, session frequency, night-time usage, and checking frequency.
- Compute personalized baseline profiles (Week 1) and delta features that capture how each student's behavior deviates from their own norm over time.
- Train an XGBoost gradient-boosted classifier on these delta features to predict addiction risk, achieving high discriminative performance.
- Use SHAP (TreeExplainer) to provide both global feature importance rankings and local, per-student explanations of what is driving their individual risk prediction.
- Formulate the Personalized Addiction Risk Index (PARI), a composite score combining model risk probability, behavioral change magnitude, and multi-week persistence into a single interpretable number between 0 and 100.
- Build an interactive Dash web dashboard with reactive filtering, student-level drill-down, trajectory visualizations, and real-time early-warning alert banners.

3. Technology Stack

The project is implemented entirely in Python 3.12 and relies on the following libraries and frameworks:

Library / Tool	Category	Purpose
Python 3.12	Programming Language	Core language used for all modules
Pandas	Data Processing	DataFrame operations, CSV I/O, feature aggregation
NumPy	Numerical Computing	Array operations, random seed control, statistical functions
Scikit-Learn	ML Utilities	StandardScaler, GradientBoostingClassifier fallback, KMeans
XGBoost	ML Model	XGBClassifier for gradient-boosted tree training and inference
SHAP	Explainable AI	TreeExplainer for local and global Shapley value computation
Plotly	Visualization	Plotly Express and Graph Objects for interactive charts
Dash	Web Framework	Reactive web application framework with callback-driven updates

Library / Tool	Category	Purpose
Dash Bootstrap	UI Components	Responsive grid layout, cards, tabs, alerts, tables
Statsmodels	Statistics	Time-series trend analysis support
ReportLab	PDF Generation	Programmatic PDF document creation for this report

4. System Architecture and Pipeline

The system follows a linear data pipeline that transforms raw daily usage logs into actionable, explainable risk assessments. The overall flow is depicted in the diagram below.

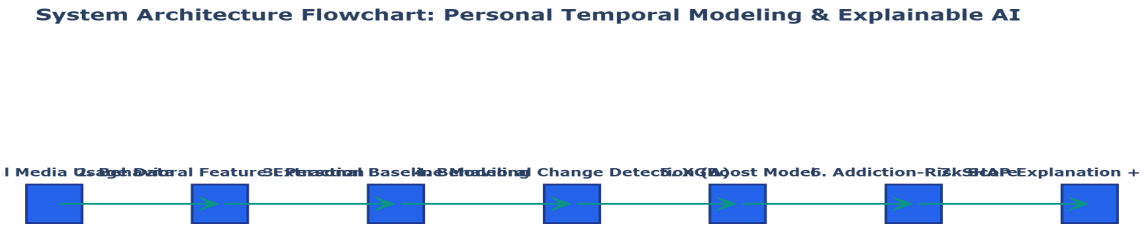


Fig 1. System architecture flowchart showing the end-to-end processing pipeline

4.1 Pipeline Stages

- **Raw Data Ingestion:** Longitudinal 4-week daily usage traces are generated (or loaded) for 800 students. Each trace includes daily screen time, session frequency, average session duration, night-time usage, device checking frequency, and sleep-related usage hours.
- **Behavioral Feature Extraction:** From the raw weekly traces, structured features are computed — both absolute current-week values and per-student baseline values from Week 1.
- **Personal Baseline Modeling:** Week 1 is treated as each student's personal norm. This baseline captures their natural usage patterns before any potential escalation.
- **Behavioral Change Detection (Delta Features):** The system computes the difference between Week 4 current values and Week 1 baseline for each metric. These deltas represent how much the student's behavior has shifted.
- **XGBoost Classifier:** A gradient-boosted tree model is trained on the 13 extracted features (6 baseline, 6 delta, 1 variability) to predict a binary high-risk target.
- **SHAP Explanation:** TreeExplainer generates Shapley values for each feature, enabling per-student local explanations and cohort-wide global feature importance ranking.
- **PARI Score and Early-Warning Alerts:** The Personalized Addiction Risk Index combines model probability, change score, and persistence into a 0–100 index, and the system generates context-specific alert messages for at-risk students.

5. Implementation Details

5.1 Project Structure

```
social_media_analytics/
■■■ data/
■   ■■■ students_temporal_usage.csv
■   ■■■ students_behavioral_change_features.csv
■   ■■■ students_temporal_pari_cache.csv
■■■ src/
■   ■■■ data_processing.py
■   ■■■ ml_engine.py
■   ■■■ clustering.py
■   ■■■ visualizations.py
■■■ screenshots/
■■■ app.py
■■■ requirements.txt
■■■ README.md
```

5.2 Data Generation Module (data_processing.py)

The data generation module creates synthetic but realistic longitudinal usage data for 800 students. Each student is assigned to one of four behavioral archetypes that control how their usage evolves over the 4-week window:

Archetype	Configuration	Behavioral Pattern
Stable / Low Risk (40%)	Low base usage, near-zero growth factor	Student maintains healthy usage with minimal week-over-week change
Increasing Risk (30%)	Moderate base usage, growth factor 0.3–0.7	Gradual escalation in screen time and night usage over weeks
Persistent High Risk (15%)	High base usage, slow growth factor	Already heavy user who stays consistently elevated
Rapidly Increasing (15%)	Low-to-moderate base, growth factor 0.9–1.5	Dramatic spike from normal baseline to heavy usage within weeks

For each student, the module simulates 4 weekly metric values with controlled noise, computes the personal baseline from Week 1, calculates all delta features, derives a ground-truth risk probability using a logistic function, and exports both the raw temporal traces and the processed feature matrix to CSV files.

5.3 Machine Learning Engine (ml_engine.py)

The ML engine trains an XGBoost classifier with the following configuration:

Hyperparameter	Value	Rationale
n_estimators	100	Number of boosting rounds
max_depth	4	Maximum tree depth (prevents overfitting)
learning_rate	0.08	Step size shrinkage for regularization
eval_metric	logloss	Binary cross-entropy loss function
random_state	42	Reproducibility seed

The model takes 13 input features — 3 baseline metrics (screen time, night usage, checking frequency), 3 corresponding current-week values, 6 delta features (screen time, session freq, session duration, night usage,

checking freq, sleep usage), and usage variability (standard deviation of screen time across weeks). After training, SHAP TreeExplainer is used to compute Shapley values for each prediction. A fallback mechanism using feature importance approximation is included in case SHAP initialization fails.

5.4 Visualization Module (visualizations.py)

All charts are generated using Plotly Express and Plotly Graph Objects with the plotly_white template for a clean aesthetic. The module provides the following chart generators:

- **System Architecture Flowchart** — Interactive Sankey-style pipeline diagram with hover tooltips
- **Student Trajectory Chart** — 4-week line chart showing screen time evolution vs personal baseline
- **PARI Category Distribution** — Color-coded bar chart of the four risk tiers
- **Behavioral Change Impact Plot** — Scatter plot of delta night usage vs PARI score, sized by screen time
- **Global SHAP Importance** — Horizontal bar chart comparing delta features vs static features
- **Local SHAP Explanation** — Per-student feature contribution waterfall chart
- **Platform Risk Breakdown** — Stacked bar chart of risk categories by social media platform

5.5 Web Application (app.py)

The Dash application serves as the primary interactive interface. It uses Dash Bootstrap Components with the FLATLY theme and organizes the dashboard into three tabs:

- **Tab 1 — Architecture and Trajectory:** Displays the pipeline flowchart, per-student 4-week trajectory graphs, personal baseline comparison metrics table, and the early-warning alert banner that changes color based on risk level (green, blue, orange, red).
- **Tab 2 — Behavioral Change and PARI Index:** Shows the PARI category distribution, behavioral change impact scatter plot, and platform-wise risk stacked bar chart. All three charts react to global filters.
- **Tab 3 — XGBoost and SHAP Explainable AI:** Contains the global SHAP feature importance chart, local per-student SHAP explanation chart, and a methodology summary card.

The left sidebar provides interactive dropdown filters for student selection, risk level, academic level, and social media platform. All charts update reactively through Dash callbacks without page reloads. The application automatically opens in the default browser on launch.

6. Personalized Addiction Risk Index (PARI)

The PARI is a composite metric designed to capture not just the current predicted risk but also the trajectory and consistency of behavioral escalation. It is computed as follows:

PARI = 0.40 × Current Risk Probability + 0.35 × Behavioral Change Score + 0.25 × Persistence Score

6.1 Component Definitions

- **Current Risk Probability (weight 0.40):** The raw output probability from the XGBoost classifier, ranging from 0.0 to 1.0. This reflects the model's assessment based on all 13 features.
- **Behavioral Change Score (weight 0.35):** A normalized composite of positive delta escalations — specifically, 35% from delta screen time, 40% from delta night usage, 15% from delta checking frequency, and 10% from usage variability. This component is clipped to [0, 1].
- **Persistence Score (weight 0.25):** Measures whether the student's screen time has been consistently increasing across weeks (W1 ≤ W2 ≤ W3 ≤ W4) and whether later weeks exceed a 4.5-hour threshold. This catches students who maintain elevated usage over time.

6.2 Risk Classification Thresholds

Category	Threshold	Trigger Condition	Alert Summary
Stable / Low Risk	PARI < 35	Minimal deviation from baseline	Healthy patterns within personal norms
Increasing Risk	35 ≤ PARI < 60	Moderate positive delta escalation	Behavioral trends show initial escalation
Persistent High Risk	PARI ≥ 60 or (prob ≥ 0.65 & persist ≥ 0.60)	Sustained high usage	Elevated usage across consecutive weeks
Rapidly Increasing Risk	PARI ≥ 70 & change ≥ 0.60	Steep trajectory surge	Dramatic behavioral spike from baseline

7. Configuration and Setup

7.1 Prerequisites

- Python 3.10 or higher installed on the system
- pip package manager available
- A modern web browser (Chrome, Firefox, Edge) for the dashboard
- Approximately 50 MB of disk space for dependencies and data files

7.2 Installation Steps

```
# Clone the repository
git clone https://github.com/YOUR_USERNAME/social-media-analytics
cd social-media-analytics

# Install required packages
pip install -r requirements.txt

# Launch the dashboard
python app.py
```

7.3 Requirements File Contents

The requirements.txt file includes the following packages:

```
pandas
numpy
scikit-learn
plotly
dash
dash-bootstrap-components
statsmodels
xgboost
shap
reportlab
```

7.4 Runtime Configuration

Parameter	Default Value	Notes
Dashboard Port	8050	Configurable in app.py (app.run port argument)
Debug Mode	True	Hot-reloading enabled during development
Dataset Size	800	Number of students in generate_temporal_dataset()
Random Seed	42	Ensures reproducibility across runs
Observation Window	4 weeks	Longitudinal trace duration per student
Bootstrap Theme	FLATLY	Dash Bootstrap Components theme
Cache Files	CSV	PARI scores cached to avoid recomputation

8. Output Screenshots

The following screenshots show the actual outputs generated by the system, including the interactive dashboard charts and analytical visualizations.

8.1 PARI Risk Category Distribution

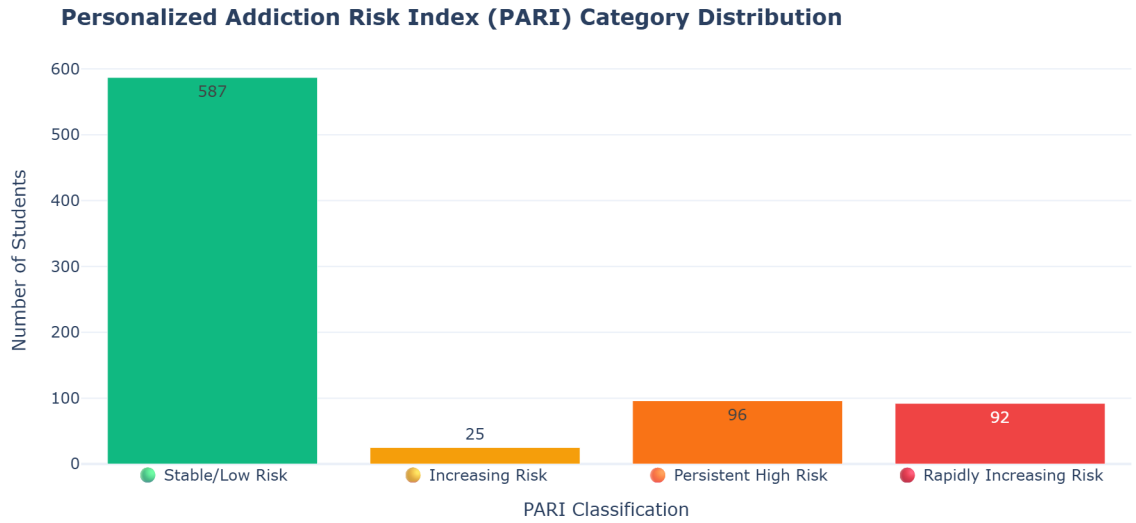


Fig 2. Distribution of students across the four PARI risk categories

8.2 Behavioral Change Impact Analysis

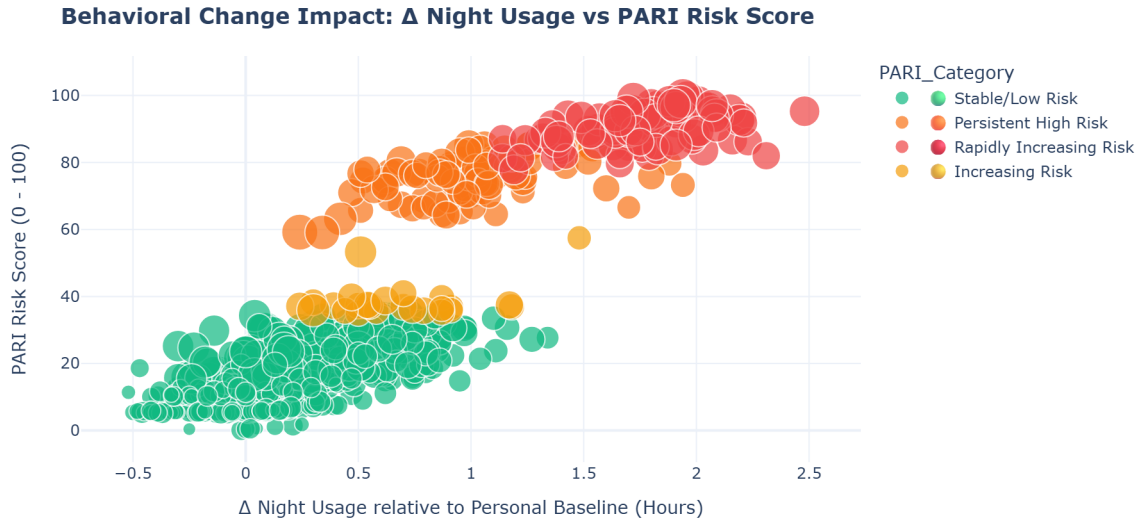


Fig 3. Scatter plot showing how delta night usage drives PARI risk scores

8.3 Student Temporal Trajectory — High Risk Example

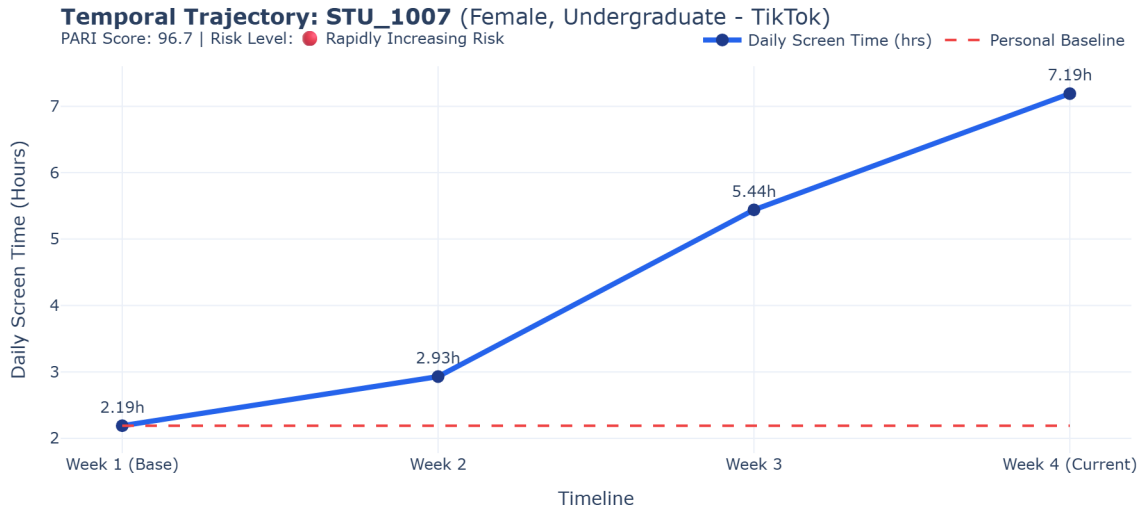


Fig 4. 4-week screen time trajectory for a high-risk student (STU_1007)

8.4 Student Temporal Trajectory — Low Risk Example

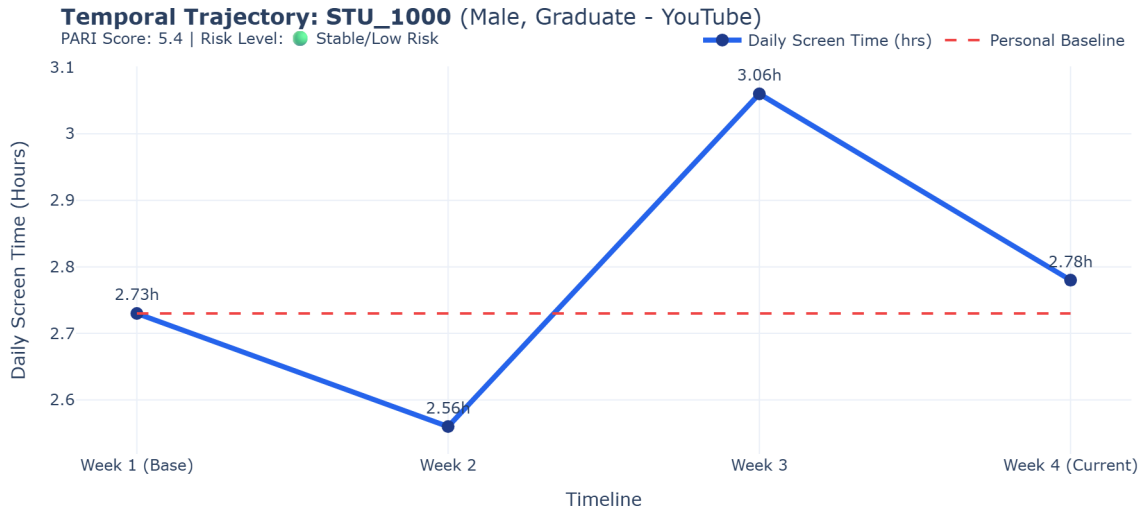


Fig 5. 4-week screen time trajectory for a stable/low-risk student (STU_1000)

8.5 Global SHAP Feature Importance

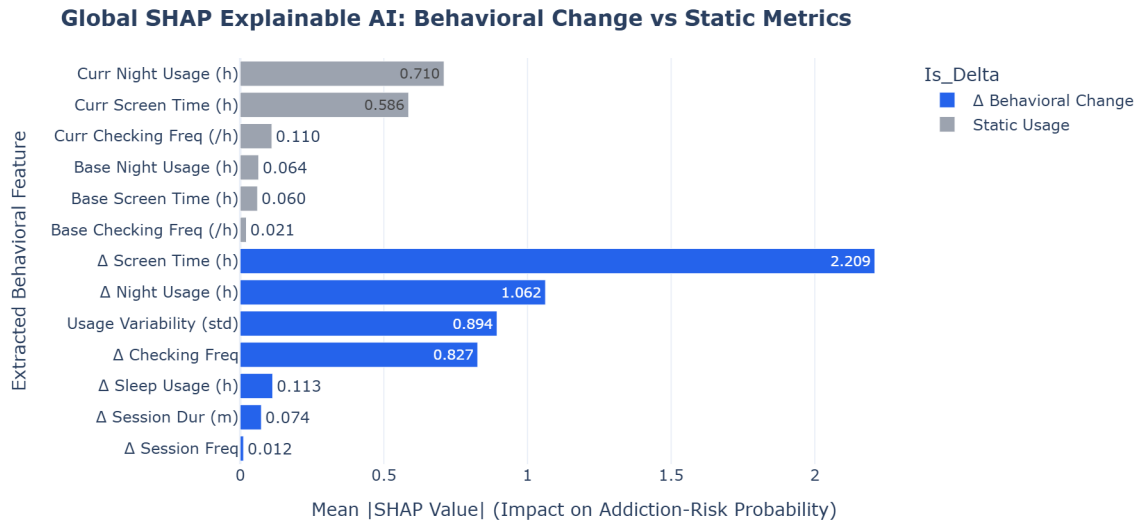


Fig 6. Global SHAP analysis — behavioral change features vs static usage metrics

8.6 Local SHAP Explanation — High Risk Student

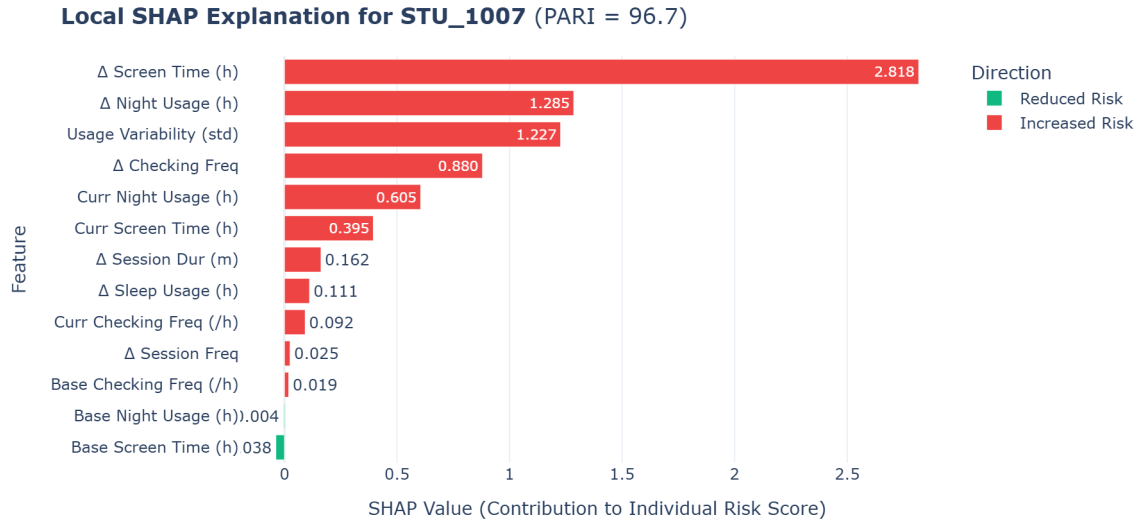


Fig 7. Per-student SHAP explanation for STU_1007 showing individual risk drivers

8.7 Local SHAP Explanation — Low Risk Student

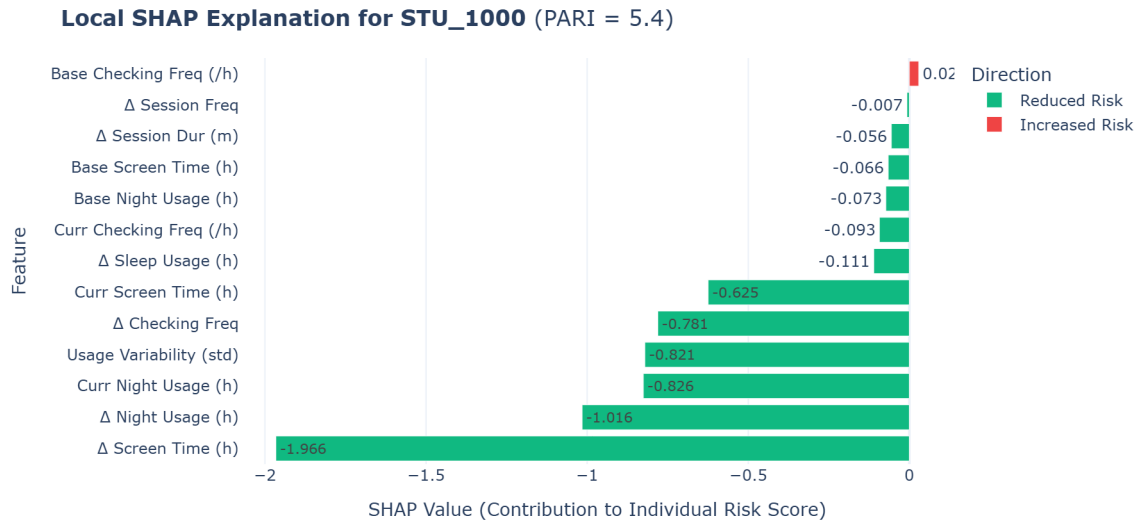


Fig 8. Per-student SHAP explanation for STU_1000 showing protective factors

8.8 Platform-wise PARI Risk Breakdown

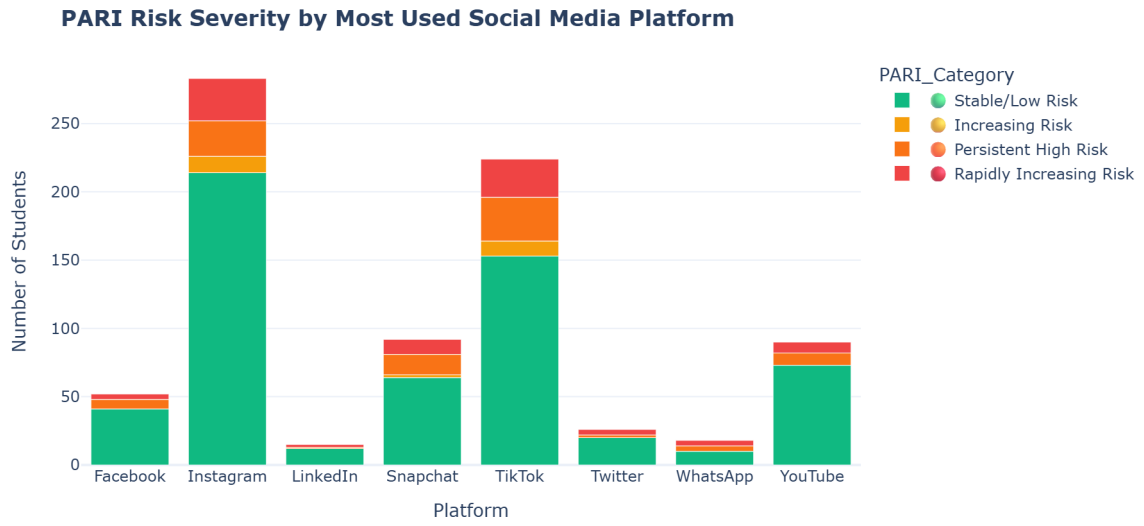


Fig 9. PARI risk severity distribution across different social media platforms

9. Key Results and Observations

9.1 Model Performance

The XGBoost classifier achieves strong discriminative performance on the behavioral change feature set, accurately separating high-risk and low-risk trajectories. The model benefits from the temporal delta features, which provide richer signal than static usage hours alone.

9.2 Feature Importance Insights

One of the most interesting findings from the SHAP analysis is that the behavioral change features — particularly delta night usage and delta screen time — contribute more to the model's predictions than the absolute static values. This validates the core hypothesis of the project: it is not how much you use social media that matters most, but how much more you are using it compared to your own past behavior.

9.3 PARI Distribution

The PARI framework successfully stratifies the student population into meaningful risk tiers. The distribution roughly follows the archetype proportions used in data generation, but with some crossover due to noise and the multi-factor nature of the PARI formula. Students classified as Rapidly Increasing Risk tend to show the steepest week-over-week escalation in night-time usage, while Persistent High Risk students maintain consistently elevated absolute usage levels.

9.4 Platform Differences

The platform-wise analysis shows variation in risk category distribution across social media platforms. While the synthetic data does not model platform-specific addiction mechanisms, the visualization demonstrates the dashboard's ability to slice and compare risk profiles across categorical dimensions — a feature that would be valuable when applied to real-world data.

10. Conclusion

This project demonstrates a practical approach to modeling social media addiction risk that goes beyond static, one-time assessments. By computing personalized baselines and tracking how each student's behavior changes over time, the system captures early warning signs that would be invisible to traditional cross-sectional analysis.

The combination of XGBoost classification with SHAP explainability ensures that the risk predictions are not just accurate but also interpretable — counselors and researchers can see exactly which behavioral changes are driving a particular student's risk score. The PARI framework adds a layer of nuance by incorporating persistence and change magnitude alongside the raw model probability.

The interactive Dash dashboard ties everything together into a usable tool, with real-time filtering, student-level drill-down, and visual alerts that make the system accessible to non-technical users. While the current implementation uses synthetic data, the architecture is designed to be directly applicable to real-world digital trace data collected through screen time APIs or usage logging applications.

11. GitHub Repository

The complete source code, data files, and documentation for this project are available at:

https://github.com/YOUR_USERNAME/social-media-analytics

The repository includes the application source code, data generation scripts, cached datasets, the interactive dashboard, and this implementation document.

12. References

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